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Department of Computer Science & Engineering

Mid-Term Examination September - 2025

Subject Name/ Code: CSE -3001 **dr. neeta anna eapen**
Time Allotted: 90 min

MM:20

NOTE: 1. All questions are compulsory.
2. Mark answer number clearly. Answers without precision receive NO credit.

✓ 1. A	✓ Explain what makes a good algorithm. List and describe at least 3 essential characteristics / properties.	3
✓ B.	✓ An algorithm takes $5n^2 + 3n + 7$ steps for an input size n . Write its time complexity using Big-O notation and justify your answer.	2
✓ 2. A	✓ Solve the recurrence $T(n) = T(n-1) + O(1)$ using substitution method (Guess and prove by induction) and state the final time complexity.	3
✓ B.	✓ Solve the recurrence $T(n) = 4T(n/2) + n^2$ using Master's method and state the final time complexity.	2
✓ 3. A	✓ For the "Optimal Storage on Tapes" problem, explain why storing programs in non-decreasing order of length minimizes the average retrieval time. Include a brief explanation using an example.	3
✓ B.	✓ Is the following statement true or false? Justify your answer briefly: "If $f(n) = O(g(n))$, then $2^{f(n)} = O(2^{g(n)})$ always holds."	2
✓ 4. A.	✓ Write down the Algorithm for insertion of an element into a max-heap. Include a brief explanation using an example.	3
✓ B.	✓ Given the following weighted undirected graph, demonstrate the steps for obtaining the Minimum Spanning Tree (MST) using Kruskal's algorithm. Graph: Vertices = {A, B, C, D, E} Edges and Edge weights: A - B: 1, A - C: 3, B - C: 3, B - D: 6, C - D: 4, C - E: 2, D - E: 5	2

All the Best